

KYLE BINDER

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APPOINTMENTS

Louisiana State University <i>RTG Postdoctoral Researcher</i>	2025 – Present
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EDUCATION

The Ohio State University <i>M.S. & Ph.D. in Mathematics</i>	2019 – 2025 Advisor: Eric Katz
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Baylor University <i>B.A. in Mathematics and Linguistics (University Scholars)</i>	2015 – 2019
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RESEARCH INTERESTS

Combinatorial algebraic geometry, combinatorics, toric varieties, matroids, tropical geometry.

PAPERS AND PREPRINTS

1. *Representations on the Top-Nonvanishing Weight Cohomology of a Smooth Toric Variety*, in preparation. (~20 pp., expected October 2026)
2. *Augmented Singular Cohomology, Uniform Matroids, and Real-Rootedness* (with Lorenzo Vecchi), preprint, [arXiv:2609.15946](https://arxiv.org/abs/2609.15946). 27 pp.
3. *The Singular Cohomology Ring of a Uniform Matroid: Combinatorics and Lefschetz Properties*, submitted, [arXiv:2606.07490](https://arxiv.org/abs/2606.07490). 21 pp.
4. *The Singular Cohomology Ring of a Matroid*, submitted, [arXiv:2412.05732](https://arxiv.org/abs/2412.05732). 35 pp.
5. *The Unipotent Tropical Fundamental Group* (with Eric Katz), *J. London Math. Soc.* (2026), 114:e70701. <https://doi.org/10.1112/jlms.70701>. 55 pp.
6. *Limit Sets and Internal Transitivity in Free Group Actions* (with Jonathan Meddaugh), preprint, [arXiv:1908.07382](https://arxiv.org/abs/1908.07382). 17 pp.
7. *On 2-coloring the integers to forbid a distance and monochromatic translates of a triangle*, *Geombinatorics* **27** (2018), no. 4, 137–146.

TALKS

<i>The Singular Cohomology Ring of a Matroid</i> University of Toronto Combinatorics Seminar	November 2026
<i>The Singular Cohomology Ring of a Matroid</i> Brown Combinatorics Seminar	November 2026
<i>The Singular Cohomology Ring of a Matroid</i> Combinatorics & Geometry BLT Seminar	September 2026
<i>A Tropical Approach to the Unipotent Fundamental Group of a Hyperplane Arrangement Complement</i> Hyperplane Arrangements and Polytopes Special Session AMS Southeastern Sectional Meeting	October 2025

<i>Cohomology of Toric Varieties from Matroids</i> LSU Geometry and Topology Seminar	September 2025
<i>The Tropical Unipotent Fundamental Group</i> 5 Minute Talk at Tropical Geometry: Moduli Spaces and Matroids Workshop	October 2024
<i>Higher Tor Groups of Matroids</i> 5 Minute Talk at Workshop of the Kentucky–Ohio Algebra Alliance (KOALA)	May 2024
<i>Higher Tor Groups of Matroids</i> OSU Combinatorics and Probability Seminar	April 2024
<i>Higher Tor Groups of Stanley–Reisner Rings of Matroids</i> 5 Minute Talk at Workshop of the Kentucky–Ohio Algebra Alliance (KOALA)	May 2023
<i>The Tropical Unipotent Fundamental Group</i> 20 Minute Talk at Workshop on Matroids and Tropical Combinatorics (MaTroCom)	January 2023
<i>A Survey of Tropical Cohomology</i> OSU Algebraic Geometry Seminar	September 2022
<i>The Geometries of Hyperplane Arrangement Complements</i> OSU Student Algebraic Geometry Seminar	April 2022
<i>p-adic Integration on Hyperelliptic Curves</i> OSU Student Algebraic Geometry Seminar	February 2021
POSTERS	
<i>The Singular Cohomology Ring of a Matroid</i> Combinatorics at the Confluence	July 2026
<i>The Singular Cohomology Ring of a Matroid</i> Texas Algebraic Geometry Symposium	April 2026
<i>The Singular Cohomology Ring of a Matroid</i> Arizona–New Mexico Symposium on Commutative Algebra	December 2025
<i>What is the Cohomology of a Non-Compact Toric Variety?</i> Kentucky, Ohio, Indiana (KOI) Combinatorics Lectures	December 2023
TEACHING	
LSU Math 4023 – Applied Algebra <i>Instructor</i>	(Expected) Spring 2027
LSU Math 2065 – Elementary Differential Equations <i>Instructor</i>	(Expected) Spring 2027
LSU Math 4997 – Vertically Integrated Research: Combinatorial Topology <i>Instructor</i>	Fall 2026
LSU Math 1550 – Calculus 1 <i>Instructor</i>	Fall 2025
OSU MATH 1131 – Business Calculus <i>Recitation Instructor</i>	Spring 2024
OSU MATH 1131 – Business Calculus <i>Recitation Instructor</i>	Autumn 2022

OSU MATH 1148 – College Algebra
Recitation Instructor

Autumn 2021

OSU MATH 1148 – College Algebra
Recitation Instructor; Virtual (Zoom)

Autumn 2020

MENTORSHIP

Mentor for Undergraduate/Graduate Research Project
Project on h -vectors of matroids (2 undergraduate/2 graduate students)

Fall 2026

Directed Reading Program (DRP) Mentor

2021 – Present

1. “The Mathematics of Chip-Firing” with E.R. (Fall 2026)
2. “The Mathematics of Chip-Firing” with C.B. (Fall 2023)
3. “Combinatorial Optimization” with G.M. (Spring 2023)
4. “An Invitation to Applied Category Theory” with J.W. (Spring 2022)
5. “Quaternions for Computer Graphics” with S.B. (Fall 2021)
6. “Basic Concepts of Algebraic Topology” with D.L. (Spring 2021)

SERVICE AND ORGANIZATION

Special Session on Algebraic and Geometric Aspects of Matroid Theory
at AMS Central Sectional Meeting
Co-organizer

October 2026

LSU Geometry and Topology Seminar
Co-organizer

2025 – Present

OSU Directed Reading Program (DRP)
Co-organizer

2021 – 2025

Kentucky, Ohio, Indiana (KOI) Combinatorics Lectures
Local Organizer

December 2023

HONORS AND AWARDS

OSU Distinguished University Fellowship

2019 – 2020, 2024 – 2025

Tibor Radó Research Semester

Spring 2021

Phi Beta Kappa

Fall 2018